

Smartphone Application for the Deaf and the Deaf Caring Community

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Abstract—In life, various challenges and problems faced by deaf people such as communication skills and other problems, including emotional, mental, and societal development. However, technology is needed that can help the process. This Smart Application try to build communication between the deaf to share thoughts in group chats, including getting information about the deaf, so that the communication learning process for the deaf becomes easier. The proposed model is using Unified Modeling Languages (UML) diagrams, such as case diagrams for the current proposed idea process and class diagrams for relational tables or database stores. In addition, the User interface (UI) is implemented in this research work ,Personal Home Pages (PHP) as a web server programming and MySQL database as an open-source database.

Keywords—Application for Deaf People, Application for Special Needs Person, Technology for Deaf People, Application for Hearing Impairment, Deaf studies.

I. INTRODUCTION

Every December 3, international disability day is celebrated in international level, giving the highest appreciation to friends, brothers, and sisters with disabilities. In general, disability is divided into four: physical disability, sensory disability, mental disability, and intellectual disability. Specifically, this study, limits deafness to one of the many human limitations included in the category of sensory disability, which is a limitation of five senses. This type of disability includes, among others, speech impaired, deaf, and blind. Mainly people who are deaf and speech impaired communicate using sign language, where deaf people are disabilities that interfere with the human hearing system and make them unable to hear the sounds around them, while mute is a disability that makes the person unable to speak freely.. The only thing that distinguishes them from ordinary people is the way of communication, and sign language is one of the ways to communicate, namely between the mute and deaf, so that one can easily live like ordinary people.

Not many previous studies have written publications on the use of technology for the deaf, and this is evidenced by the results of searches conducted using Google Scholar by searching words "technology deaf people" on the paper's

title, which found only 54 results for all years (https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=allintitle%3A+technology+deaf+people&btnG=).

Moreover, there are only 12 publications in the last five years since the 2021 search (https://scholar.google.com/scholar?as_ylo=2017&q=allintitle:+technology+deaf+people&hl=en&as_sdt=0.5).

God created humans with strengths and weaknesses from each other, and one must find solutions to problems and shortcomings in this life. For people who are deaf, it is one of the limitations of humans, both acquired from birth or acquired due to one or two things due to an unavoidable accident. Because of this limited sense of hearing, one cannot use spoken language and only communicate through sign or written language.

Actions and behaviors that make them even more desperate. At first glance, this deaf disability appears as a limitation that is not too problematic, but it often causes many problems for deaf people considering the limited communication that makes it difficult for them to socialize with their parents, children, husband, wife, spouse, and family. Especially for deaf children, who feel their world and lonely, it is impossible to make them mentally weak given their limitations and weaknesses coupled with regular game friends who subconsciously bully and hurt them well. In addition, in terms of getting an education, it is also a problem where deaf students can only communicate with the teaching of a teacher or lecturer in one dimension, namely through writing, drawing, or animation media.

Despite these weaknesses, one should be grateful that the deaf can still see the beauty of this world created by God Almighty. Moreover, deaf persons can still use their mouths to communicate even though it is only in one direction, and at least their tongues and lips are still functioning perfectly, and this is something to be grateful for as His creatures who are always filled with gratitude for His creations that make their hearts always calm and optimistic face the future, always in His great plan and providence.

Seeing the limitations experienced by our fellow deaf people, especially in terms of communication and socialization with those around them as well as in terms of problems in obtaining limited education because of their deafness, so a prototype of this application should be

designed to at least help them communicate and socialize with each other. In addition, due to the concerns of families who have family members who are deaf, it is also essential that this application is made in the future so they can learn sign language to build socialization and communication with those who are deaf. So that in the end, the dream of a complete family, regardless of limitations, one of the family members does not dampen their hearts and makes this heart always grateful and grateful to its creator, the almighty god.

II. PREVIOUS AND CURRENT IMPLEMENTATION

In this section, we explore several previous research papers on a similar topic because we find there is an implementation of Augmented Reality (AR) technology to help the problem of deaf people. A literature review paper in the area of application for the deaf is also discussed here and discusses several papers describing the process of translating everyday languages into their country's sign language, such as Colombian sign language, Portuguese sign language, Brazilian sign language, Malaysian sign language, and American sign language. An experiment in designing a panic button application was carried out to help deaf friends, and in addition, an application was developed to provide sex education for the deaf. In addition, an application was developed for the deaf to understand decision-making in forensic activities carried out by doctors. Finally, an application for deaf Muslim students was developed to study the holy book Al-Quran in Malaysia. Besides that, interestingly, video games were developed specifically for deaf students to learn mathematics in fun and not boring way.

Applications that students can learn on smartphones are intended not only for deaf people but also for people or families with deaf family members to learn and practice communicating with deaf people. The first paper discussed an Augmented Reality (AR) application implemented on Android to display three-dimensional animated hand shapes to learn hand sign language for the deaf. The application of this application is made by capturing an image of a letter using a smartphone camera, and this application will automatically run an image processing algorithm to recognize the letters read from the camera and display a three-dimensional hand image that corresponds to the letters scanned by the smartphone camera[1].

Next, the following paper made an application was developed by applying augmented reality (AR) technology developed for the exhibition space at a museum at the Federal University of Rio de Janeiro, Brazil, where AR can provide detailed explanations to museum visitors who have hearing problems or better hearing known as deaf. The AR application is described silently, considering that the user is deaf, so there is more information in the form of writing, images, and animations to make it easier for the deaf to understand the objects or objects on display in the museum room[2].

Moreover, an application called the panic button is proposed to help and monitor someone who requires help or assistance and indeed to monitor especially for people with special needs and especially in this case mainly for deaf people. In this study, a user interface redesign was carried out to build a panic button system that was more in line with

the needs of people with deafness, which was carried out using the User-centered design method [3].

Interestingly, a study was conducted to design and evaluate the usability of a web-based application in order to promote sexual health education for young people, especially young people with deafness, in Mexico, involving seven members of the deaf community in Mexico. ProTGTism was developed, which has five modules of healthy sex education, and this application was evaluated by 13 young deaf people who consist of seven men and six women aged between 18 to 29 years. The evaluation described the developed application as easy to use, informative, and interesting for the students and participants [4].

Moreover, a translation application for deaf people is implemented with the aim of helping deaf people when faced with an agreement to carry out a medical approval activity that involves a doctor and requires full awareness and understanding of the patient about the causes and consequences of carrying out a forensic activity. The main reason is that there is a lack of communication or lack of understanding caused by language and communication with deaf people in a situation of approval of forensic activities, so multiple translation communication is needed in this case so that deaf people can understand the message in detail as a whole so that the decision on forensic activities carried out by personnel medical is really something they understand[5].

Furthermore, subsequent research conducted a literature review on the Sign Language Recognition System, which recognizes an image in the form of letters and is matched with sign language symbols for a letter. As described by this paper, in terms of getting data, one of them is a camera or Microsoft Kinect, which is used to capture images of characters, and besides that, several other technologies are used to capture the image data. In terms of image data processing, there are several algorithms used based on this survey paper, namely algorithms such as Hidden Markov Model (HMM), Fuzzy K means, Convolutional Neural Network (CNN), Artificial Neural Network (ANN), Naive Bayes Classifier (NBC), MultiLayer Perceptron (MLP), Self Organizing Map (SOM), Self Organizing Feature Map (SOFM), Simple Recurrent Network (SRN), Support Vector Machine (SVM), and others [6].

Additionally, researchers from Colombia in South America created an application that changes the meaning of writing from Spanish to Colombian sign language using machine learning algorithms to apply Artificial Intelligence (AI) to think like a human. This translation from Spanish to Colombian sign language not only helps the deaf but also helps the mute in understanding a text or document, especially for people with disabilities who are still in the educational stage at the age of primary, secondary, high school, or college education [7].

Besides, researchers from Portugal created a project called Educacaoacessivel.pt which aims to make a video game a serious game that can be accessed free of charge to teach mathematics lessons to the deaf by using Portuguese sign language. Making video games is intended so that these deaf students can be interested in playing games, and unconsciously, they are learning mathematics, so they learn mathematics in a fun and entertaining way. Considering that

these educational video games are intended for students with hearing impairment, the sound is almost non-existent in these games, and most of the text, images, and animations that dominate these video games make it easier for them to play these video games, which in turn allows them to learn mathematics in an easy way and more fun[8].

Further, researchers from Brazil have conducted research specifically for deaf children who will enter universities in Brazil and are devoted to implementing Brazilian sign language courses. This study applies the results of their research in the form of eight hypotheses obtained from literature review, interviews, observations, forums group discussions and includes listening to competent experts in the field of deaf studies[9].

Likewise, a researcher from America in his thesis implemented a sign language translation application, especially in this study American sign language for deaf people in order to communicate better with people around them when most deaf people are always limited in their academic activities and are always assisted by the people around him to be able to access the sounds they could not hear. This voicer application is implemented with Electromyography (EMG) sensing technology, where every movement of the human body can be detected, such as arm and fingertip movements translated as part of American sign language communication. Of course, using camera captures stored in images translated into language form a sign that a deaf person can understand[10].

Lastly, a year-end project by a student in Malaysia to create a mobile application for deaf children who are Muslim to study the Holy Qur'an. These deaf students who are Muslim have difficulty listening to the directions and teachings of the educators as well as educators who have difficulty teaching deaf students to learn and understand the holy verses of the Qur'an. This proposed mobile application helps teachers to educate deaf students by using additions and colors to read the Qur'an, and the creation of this application is based on previous studies and refines it to make it easier for deaf students to understand and learn the Qur'an. Learning the Qur'an for deaf Muslim students is not only a learning requirement but also a way of life for Muslim students to become better in morals and life in a pluralistic society[11].

Another paper researched deaf children who will enter universities in Brazil and are devoted to implementing Brazilian sign language courses. This study applies the results of their research in the form of eight hypotheses obtained from literature review, interviews, observations, forums group discussions and includes listening to competent experts in the field of deaf studies[12]. A project was created using a video game, a serious game that can be accessed free of charge, to teach mathematics lessons to the deaf by using Portuguese sign language. Making video games is intended so that these deaf students can be interested in playing games, and unconsciously, they are learning mathematics, so they learn mathematics in a fun and entertaining way[13].

Another paper created an interactive web-based media to introduce and learn Indonesian sign language, which has proven influential in introducing it to the public, especially for people with hearing and speech impairments[14]. Additionally, an application changed the meaning of writing

from Spanish to Colombian sign language using machine learning algorithms to apply Artificial Intelligence (AI) to think like a human. This translation from Spanish to Colombian sign language not only helps the deaf but also helps the mute understand a text or document, especially for people with disabilities who are still in the educational stage[15].

A doctoral dissertation implemented an android-based application for learning Indonesian sign language, and the experimental results using this methodology succeeded in recognizing BISINDO gestures with an accuracy of around 60%-70%[16]. It is hoped that this application built can reduce the dependence of deaf people on sign language interpreters in bridging communication between deaf people and those with hearing disabilities [17]. Another doctoral dissertation conducted research and applied a virtual teacher that was used as a tool for teachers to help them simulate Indonesian sign language lessons using an animation-based application that helps students to learn sign language[18]. In addition, a dissertation discusses the application of Indonesian sign language for letters and numbers using a desktop-based application that is applied to all Special Schools in the city of Bandung, and it is proven that the use of technology in learning increases the interest in deaf students to learn Indonesian sign language [19].

Moreover, an application was developed for the introduction of Hijaiyah letters for deaf students who are Muslim and is expected to help them to be able to read the Al-Quran correctly and adequately [20]. Furthermore, a doctoral dissertation developed a web-based Indonesian sign language learning system, and it proved to be quite entertaining and complete, and the average percentage of ratings given by users was 58.54%[21].

III. PROPOSED IDEA

In this section, the proposed ideas are designed and implemented where the design process uses Unified Modeling Languages (UML) tools such as use cases and class diagrams. The use case diagram describes the proposed processes, as shown in Figure 1, while the class diagram in Figure 2 shows the database table relationships implemented using the open-source MySQL database. The application uses Personal Home Pages (PHP) as an open-source web-based server programming.

The use case diagram in Figure 1 shows the process of proposing a smartphone application for the deaf with five use case activities: registration, login, group chat, information, and the nearest location. The registration process is a requirement where the user who wants to use the application must enter the data as shown in the user class as shown in the class diagram in Figure 2. The user interface for the registration process can be seen in Figure 3(b), and input data is associated with a user class in a class diagram. Next, a registration confirmation will be sent to the registered user's email address.

This application will check the email and password entries to the user table database to check the match of the typed email and password with those in the user table, as shown in the class diagram in Figure 2 as a user class. If the email and password match the storage database, then the user will be allowed to enter the main menu of the application, as

shown in Figure 4(a), and if the email or password does not match, then there will be a message that will remind the user that they do not have access to enter the application and confirm if they do not have an account yet, they must register first. Furthermore, the login process in the use case diagram is the process after the user registers, where the user must enter his email and password, while the authentication process, where the application user interface displays, can be seen in Figure 3(a).

Next, use case group chat to show the communication process between user members to discuss all information about deaf people. As previously stated, this application can be used not only by deaf people but also by their family members. Apart from communication between deaf people, this can also educate them on how to deal with deaf people and how to interact and socialize properly and correctly with deaf people, to maintain their feelings, so they do not feel bullied by their shortcomings. Furthermore, figure 4(b) shows the application's user interface, where the user can bring up a new chat and reply to any chat and which, in the process, will use two classes such as thread and reply, as seen in the class diagram in figure 2. The thread class will create a new record every time a user submits a new chat, and so in the case of the reply class, a new record will be created every time the users submit a reply to the chat.

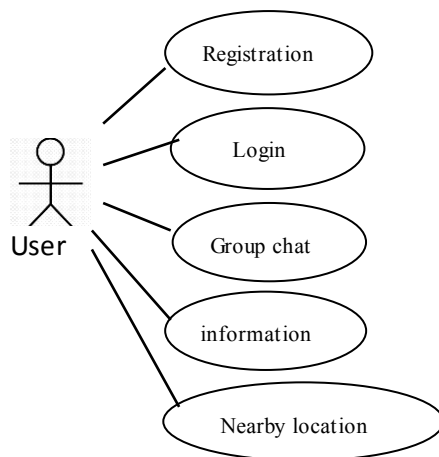


Fig. 1. Use case diagram of a smartphone application for deaf people

Meanwhile, use case information is a process that aims to show all information needs for this community where each member can send their ideas about the information and send it to the admin, who will post the ideas. This large amount of information is also demanded from the activeness of members to provide positive and relevant information. Regarding the community that cares for the deaf, it is hoped that both stories, including tips and tricks for the community caring for the deaf, will benefit both the deaf and people listed here who are also family members with a family member who is deaf. Figure 4(a) shows the user Interface for this information display.

Lastly, the Nearby location use case is a process that can monitor the location of deaf people, especially for deaf people who are underage or are still children or deaf people who are aware that they need their family and community when they are far from their family. This is very important, considering that not all deaf people know they are deaf, and deaf people cannot be seen physically in terms of their shortcomings. In addition, many bad people take advantage

and deceive someone when they find out they are disabled. This is necessary if, in one way or two, they get lost when the parents of the deaf are so that their children are more mature and can be released freely to be able to mingle with others who care and know their limitations, and it is essential to detect their location, and this is expected to mature deaf children who are still children so that they become more confident. Figure 5 shows the tracking of deaf people when they are set up via their smartphone by activating the connected Global Positioning Systems (GPS) and can actively find their current location through the activated application. However, there are still weaknesses when the smartphone runs out of power, or the application is not activated. It will be challenging to monitor and detect the location of the deaf person, and this is still a weakness of this application's application.

Meanwhile, figure 2 below shows a class diagram where there are three classes, namely thread, user, and reply, and each class is represented as a table in the database, which is implemented as three database tables using a MySQL database. For the relationship, it can be seen that starting from the user class, it has a one-to-many relationship to the thread class, which means that each user can have one or many threads, while each thread record that is formed comes from a single user. The user class also has a one-to-many relation to the reply class, meaning each user can make one or many replies while a single user creates each reply record. Meanwhile, the Thread Class has one-to-many relationships to the reply class, which means that each created thread record can have one or many replies, and each reply record formed is for a thread.

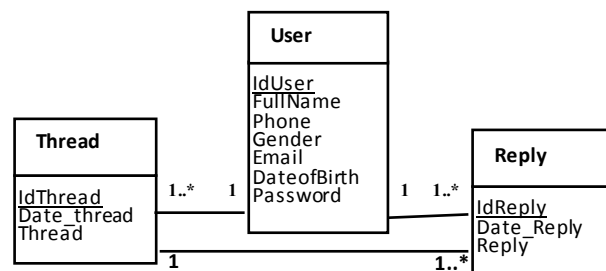


Fig. 2. Class diagram of a smartphone application for deaf people

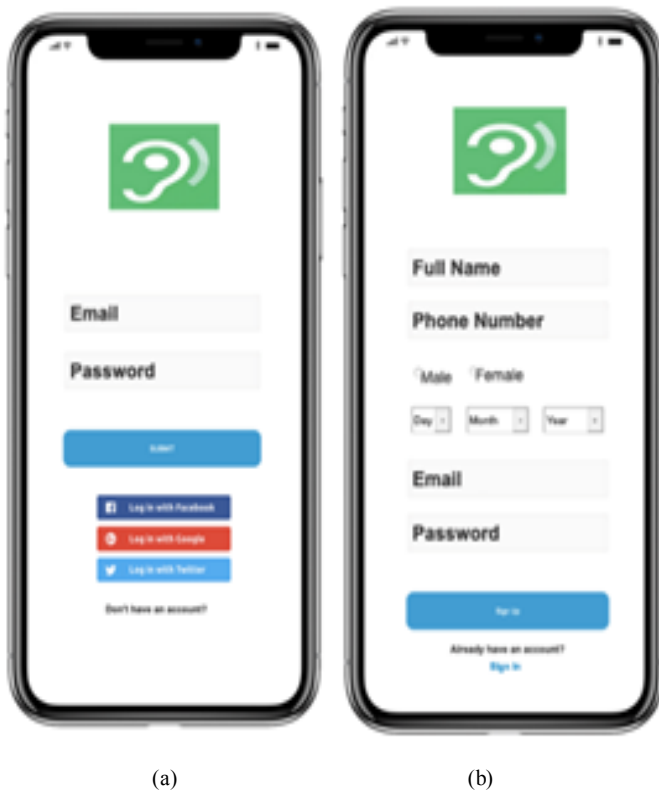


Fig. 3. (a) User interface login, (b) user interface registration

As explained in the description of the use case diagram above, the following User Interface (UI) pictures display each use case diagram activity in Figure 1. Figure 3(a) shows the UI for the login system, which is mentioned in the use case diagram as a login use case, and before doing the login process, the user must first register and enter an email and password for authentication. The system will check the input email and password and check it into the appropriate user database table, which is seen as a user class in Figure 2. Then if it matches, the user can enter the system; if not, it will be rejected and includes a warning that they do not have access and information for signing up if they do not have an account yet. Also, there will be a warning if they forget their email and password and especially for forgot password when they have to enter the email address, and admin will send the new password to the user to have a new password to log in, and then they can also update the password them again.

Figure 3(b) is a UI for registration where users have to enter their full name, phone, gender, email address, date of birth, and password, and all data will be stored in the user table, as seen in the user class in Figure 2. As seen in the use case diagram in Figure 1, the UI in Figure 3(b) is a registration use case activity.

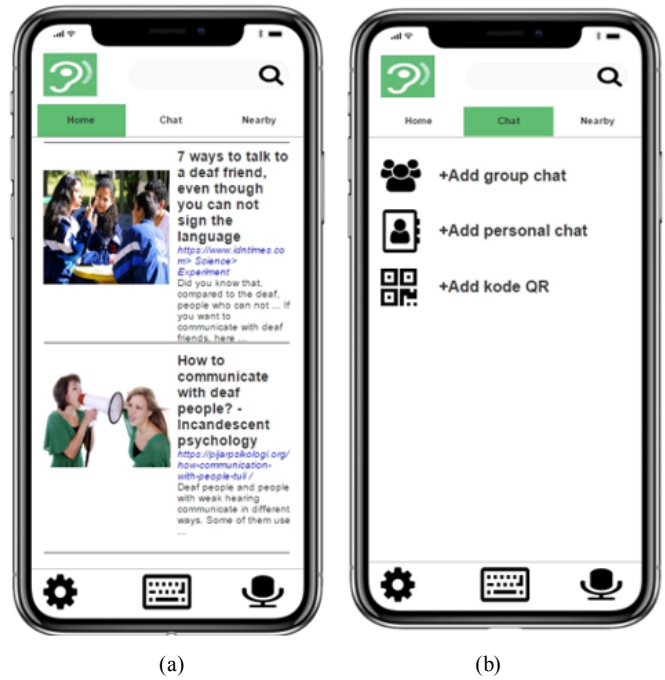


Fig. 4. (a) User Interface Information, (b) User Interface Group chat

Meanwhile, Figure 4(a) shows the information needed by the deaf and each member in this application, following what is described in the use case diagram in Figure 1 as an information use case activity. The admin will upload, update, and moderate this information delivery activity. The information obtained is also obtained from members of the users of this application which aims to educate and learn about all problems related to the deaf.

Figure 4(b) shows a UI for group chat that is useful for building communication between members registered in the use of this application, both communication between the deaf and communication with non-deaf registered in this application, which of course, is family deaf members. The UI for this group chat follows what has been shown in the group chat use case activity as shown in Figure 1, and in the process, this group chats communication will be stored in 2 database tables, such as threads and replies shown in the class diagram image in Figure 2 as a class user.

Finally, figure 5 shows the Nearest Location UI, which can find the position of the deaf, especially for deaf children who are required to activate it first via their smartphone using GPS smartphone technology. The Nearest Location UI in Figure 5 is part of a use case activity such as a nearby location, which is part of an application that aims for security purposes, especially for children with hearing impairments when they are far from their loved ones registered in this application.

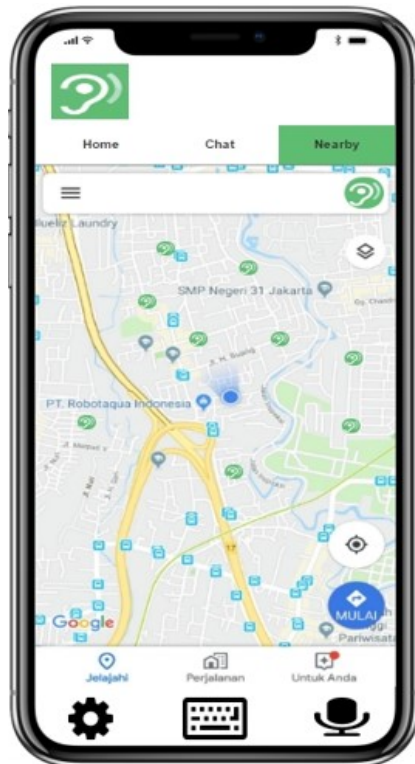


Fig. 5. User Interface Nearby location

IV. CONCLUSION

This study concludes that people who are interested in caring about their fellow persons with disabilities, especially those who are deaf that this application can provide encouragement and role models, especially for people with disabilities, for the attention given to them both from fellow persons with disabilities and their accompanying families. In addition, for families who have family members who are deaf, this provides a spirit of togetherness to come together in a community and support each other among people with disabilities.

The implementation of this application is still straightforward, and future development is still needed, such as the application of Artificial intelligence (AI) technology implementation.

In future, it is essential to apply to learn about Indonesian sign language, which uses a game approach so that it is more exciting and entertaining. In addition, image processing algorithms will also be applied, which use a camera or video that can capture a sentence and translate it automatically into Indonesian sign language. In addition, with the same technology, the voice of the sentence pronunciation using voice-to-text technology will automatically change the sentence's pronunciation to the Indonesian sign language form. In addition, Kinect sensor technology is also crucial in future applications to help deaf people use Indonesian sign language.

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